UNITED REPUBLIC OF TANZANIA

INFORM Subnational Disaster Risk

The Country Model Template: how risk is calculated, the bottleneck, and scaling it to local level

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INFORM: how the Country Model Template is built

- INFORM is an open composite index of disaster risk: one 0–10 score per area (0 = lowest, 10 = highest). Tanzania runs the regional INFORM Country Model Template (Excel, 2024 release) at 170 district-level units. INFORM publishes no national figure for a subnational model; the INFORM aggregate of all 195 councils on the workbook data is 3.8 (Hazard & Exposure 2.4, Vulnerability 3.8, Lack of Coping 6.2).
- **Risk is built from three dimensions:**
  - Hazard & Exposure: drought, flood, earthquake, cyclone, conflict … and who/what is exposed
  - Vulnerability: poverty, health, nutrition, food insecurity, displacement
  - Lack of Coping Capacity: institutions & infrastructure (DRR, governance, WASH, health, education)
- **Structure: 78 raw indicators (53 used) → 6 categories → 3 dimensions → 1 risk score, for every unit.**

How an indicator becomes a 0–10 score (the hidden-sheet method)

1. Actual value, in its natural unit
2. Denominator: ÷ population / area / GDP, or none
3. Outlier cap: Tukey IQR fence: MAX(MIN(x, Q3 + 1.5·IQR) , Q1 − 1.5·IQR)
4. Transform: None | Logarithm LN(0.001 + x) | Exponential
5. Min–max scaling: 10 × (x − Min) / (Max − Min) against a fixed reference range
6. Direction: if protective (higher = better): 10 − score
7. **Clamp to [0,10] → ROUND to 1 decimal → the 0–10 indicator**

- Every indicator, whatever its unit, passes through the same chain: Tukey outliers, log/exp transform, min–max, direction.

A few indicators: many units (why standardisation is needed)

Indicator	Unit	Dimension
Food security: insufficient food	% of population	Vulnerability
Historic drought frequency	years	Hazard & Exposure
Internally displaced people	count (people)	Vulnerability
Gross National Income per capita	US$ (thousands, log)	Coping Capacity
Health expenditure per capita	US$ per capita	Coping Capacity
Physicians density	per 10,000 population	Coping Capacity
Measles incidence	per 1,000,000 population	Vulnerability
Soil erosion	Mg / ha / yr	Hazard & Exposure
Coastal erosion	metres / year	Hazard & Exposure

%, years, counts, US$, per-1,000, per-10,000, Mg/ha/yr, m/yr: all reconciled onto one 0–10 scale.



From indicators to risk: categorisation & aggregation

- **Component = AVERAGE of its indicators where Use = "Yes" (Excel AVERAGEIFS)**
- **Category = AVERAGE of its components (arithmetic mean)**
- **Dimension = scaled GEOMEAN of its categories**
- **Risk = $\sqrt[3]{(H \times V \times LCC)}$: cube-root of the three dimensions**
  - ROUND(H^(1/3) × V^(1/3) × LCC^(1/3) , 1)
- The geometric mean matters: a unit cannot "average away" one very weak dimension: high hazard with low coping capacity still yields high risk.



The honest problem: it all lives in deeply hidden Excel sheets

- **Every reference range, outlier fence, transform and aggregation step is buried in hidden worksheets.**
- Opaque: the math is invisible to the people who must use and trust the numbers.
- Fixed unit set: built for ~170 districts; it cannot scale to 195 councils, wards or villages.
- Manual & single-file: one workbook, no version control, easy to break, hard to audit.
- Closed to local data: no clean way to add a local indicator or feed council-level values.
- No automation: no live update, no link to early warning or anticipatory action.
- **Result: the model stops at its 170 district-level units of 2024. Council, ward and village risk, where decisions are made, is out of reach.**



We rebuilt the hidden sheets as an engine: proven identical, built to scale

- The hidden-sheet formulas were reverse-engineered and reproduced exactly in open code.
- **Proven byte-for-byte against the official workbook:**
  - 8,664 / 8,664 standardised indicator cells reproduce the template exactly
  - 170 / 170 district risk scores reproduce the template exactly
- Now it scales: the same engine runs unchanged on 195 councils and 31 regions.
- Spec-driven & transparent: add, delete or amend an indicator without touching the math; fully testable.
- **The regional INFORM core is never altered: it is reproduced exactly; everything new is additive.**

Capturing real local data: the methods

- Federated collection tools: one shared Excel per sector; they enter ACTUAL values in natural units and a locked formula shows the 0–10 live.
- One accountable stakeholder per indicator: TMA (climate), GST (geology), NEMC (environment), MoH (health), NBS (statistics), MoA/MUCHALI (food), MoW (water), PMO-DMD (DRR).
- Bottom-up aggregation: village → district → region → country; finer, well-filled data refines the higher level.
- Authentic sources: NBS 2022 Census, TMA CHIRPS/ERA5, GST seismic, MoH TDHS, IPC/MUCHALI, TASAF, and satellite Earth Observation (validated before use).
- **The regional INFORM core is never altered. Institutional values enter by the INFORM method with a citable source; every other Tanzanian addition is shown in a separate Advancement layer.**

The gap: what still has to be solved

- **Subnational risk is not yet operational: decisions happen at council / ward level, but the regional INFORM model stops at 170 district-level units.**
- Local data is scattered and uneven: many key indicators exist only at region or national level.
- No sustainable pipeline: capturing, validating and updating local data is still manual.
- Earth-Observation indicators (vegetation, soil moisture, flood, landslide) need proper computation, downscaling and validation.
- No live link to action: risk is not yet wired to the emergency operations centre, early warning, or local planning & budgeting.
- Capacity & financing gaps at local-government level: tools, training and incentives to keep the data flowing.

Opportunities: research & financial support for member-state adoption

- **RESEARCH**
  - Earth-Observation computation & statistical downscaling to council / ward level
  - Validation against ground events; multi-source "exploded" indicators; climate-seasonality into early warning
- **FINANCIAL SUPPORT & ADOPTION**
  - Fund the local-data integration and the federated tools; build local-government capacity to capture & submit
  - Scale the transparent engine across SADC member states; integrate risk into EOCC, anticipatory action & local planning
- **Tanzania as the subnational-INFORM pilot: a shared, open method and data standard for the region.**



Conclusion

- The Country Model Template math is fully understood and reproduced exactly: 8,664/8,664 cells, 170/170 risks.
- It now runs at council level (195) because it is open code, not a hidden-sheet workbook.
- Earth-observation layers are shown apart from the Core as a labelled Advancement layer (12 of 25 sub-indicators filled); a composite drought index and a Sentinel-1 flood layer were withdrawn when validation failed.
- **Ask: research partnership + financing to integrate local data and support member-state adoption and local-level integration.**

